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COMPARATIVE EVALUATION OF WEAR RESISTANCE AND SURFACE ROUGHNESS OF INJECTABLE VERSUS CONVENTIONAL NANOHYBRID COMPOSITE RESINS

(In Vitro Study)

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in

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Abstract

Background. Highly filled injectable composites are marketed for use in load-bearing restorations. Independent data comparing their gravimetric wear and surface roughness after standardised toothbrush abrasion with those of a conventional nanohybrid resin remain limited.

Aim. To compare percentage weight loss and arithmetic mean roughness (Ra) of Beautifil Flow Plus X F00, G-ænial Universal Injectable and Beautifil II LS after 10 000 toothbrushing cycles.

Methods. Thirty-six discs (10 mm diameter × 1 mm thick; n = 12 per group) were subjected to 10 000 brushing cycles at a load of 2 N in a Colgate Total slurry prepared according to ISO guidance. Specimen mass was recorded before and after abrasion. Ra was measured by contact profilometry. Percentage weight loss was compared by one-way ANOVA. Ra after brushing was compared by the Kruskal–Wallis test ($\alpha = 0.05$).

Results. Mean percentage weight loss was 1.49 ± 3.14 % for Beautifil Flow Plus X F00, 0.65 ± 0.37 % for G-ænial Universal Injectable and 0.41 ± 0.25 % for Beautifil II LS (p

= 0.329). Mean Ra after brushing was 0.104 ± 0.017 µm, 0.100 ± 0.023 µm and 0.194 ± 0.050 µm respectively ($p < 0.001$). Each injectable resin was significantly smoother than Beautifil II LS. The two injectable materials did not differ from each other. One Beautifil Flow Plus X F00 disc lost approximately 11.4 % of its mass and accounts for the large standard deviation in that group.

Conclusions. Under this toothbrushing protocol, gravimetric wear did not differ significantly among the three resins. Both injectable materials remained significantly smoother than the conventional nanohybrid. Mean Ra of the injectable resins stayed below the 0.2 µm plaque-retention threshold, whereas the nanohybrid mean lay on that threshold.

Keywords. Injectable composite; nanohybrid composite; toothbrush abrasion; wear; surface roughness; giomer.

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List of abbreviations

ANOVA — analysis of variance

ISO — International Organization for Standardization

Ra — arithmetic mean roughness

RDA — relative dentine abrasivity

SD — standard deviation

S-PRG — surface pre-reacted glass-ionomer

ΔRa — change in Ra (after − before)

Chapter 1

Introduction

The clinical success of a resin composite depends, among other properties, on how the surface behaves under daily use. Wear flattens occlusal anatomy and can reduce vertical dimension. Increased roughness retains plaque, takes stain and favours secondary caries at the margin (1,2). Toothbrushing is a daily source of both problems. Dentifrice abrasive acts as a third body: the resin-rich surface layer is removed first, filler particles then stand proud or are dislodged, mass is lost and Ra rises (3,4).

Highly filled injectable resins were developed so that a syringeable material could be placed in cavities previously reserved for packable composites. Filler load is raised, particle size is reduced and silane treatment is improved until the paste still flows but no longer behaves like a classical low-filled flowable (5,6). G-ænial Universal Injectable (GC Corporation, Tokyo, Japan) and Beautifil Flow Plus X F00 (Shofu Inc., Kyoto, Japan) are marketed on that claim. Beautifil II LS (Shofu Inc.) is a sculptable low-shrinkage nanohybrid giomer and served as the conventional control in this study.

Direct comparative data on these three products, tested together under three-body toothbrush abrasion, remain scarce. Two-body chewing-simulator studies are not a substitute: they load a steatite antagonist at approximately 49 N and measure volumetric loss rather than mass loss in a toothpaste slurry (6). The present experiment examines the other wear mode — toothbrushing according to ISO guidance, at 2 N, for 10 000 strokes, conventionally taken as about one year of twice-daily brushing (3,7,8).

An Ra of 0.2 µm is the threshold above which plaque retention increases (1). Surfaces smoother than this do not retain less plaque in a clinically meaningful way. A restoration rougher than about 0.5 µm can be detected by the tongue (2). Those two values provide the clinical frame for the roughness results reported here.

1.1 Aim of the study

To compare wear resistance, expressed as percentage weight loss, and surface roughness (Ra) of two highly filled injectable composites with a conventional nanohybrid composite after standardised toothbrushing.

1.2 Objectives

1. To measure specimen mass before and after 10 000 brushing cycles and to calculate percentage and absolute weight loss.
2. To measure Ra before and after the same abrasion and to calculate ΔRa.
3. To compare the three materials using the statistical tests appropriate to each outcome.

1.3 Null hypotheses

1. There is no significant difference in percentage weight loss among Beautifil Flow Plus X F00, G-ænial Universal Injectable and Beautifil II LS.
2. There is no significant difference in Ra after toothbrushing among the same three materials.

Chapter 2

Review of literature

2.1 Wear of restorative resins

Wear is the progressive loss of substance from two surfaces in relative motion (9). In the mouth the process is mixed. Attrition is two-body contact; abrasion at non-contact sites is usually three-body, with food or toothpaste as the loose agent; corrosion and fatigue accompany both (9,10). Laboratory ranking is useful for comparing materials under controlled conditions, but it does not predict clinical lifetime.

Filler volume, size, shape, inter-particle spacing, the resin matrix, the degree of conversion and the quality of the silane layer all influence wear (11,12). Condon and Ferracane varied filler volume, silane treatment and degree of cure in an oral-wear simulator. Wear increased once filler volume fell below 48 vol% and rose linearly as the fraction of silane-treated filler was reduced. Degree of conversion moved wear in the expected direction, although the effect was smaller (12). Smaller particles and a well-coupled filler bed protect the matrix. Classical flowables, filled at about 40–55 wt%, wear faster than packable hybrids for this reason (5,13). Raising filler load and reducing particle size was the route taken with the newer injectables. G-ænial Universal Injectable is listed at 50 vol%, just above Condon's 48 vol% step. Beautifil II LS is listed at 83 wt%; its volume fraction is not stated by the manufacturer.

Toothbrushing tests follow ISO/TR 14569-1. The stated scope of that technical report is wear of artificial teeth and veneering materials, but the load range (0.5–2.5 N) and the requirement for a dentifrice meeting ISO 11609 have been adopted widely for composite screening (3,14). Ten thousand strokes are commonly treated as one year of twice-daily brushing if a patient makes about fifteen strokes per surface, twice a day (7,8). The conversion is an estimate, but it remains the figure used in most toothbrushing papers.

Wear may be recorded as volume, profile depth or mass. Gravimetric loss is simple and repeatable when the balance is sufficiently sensitive and the specimen is dry; it does not describe the shape of the worn surface. Volumetric loss from a scan does. The two quantities answer different questions (10,15).

2.2 Surface roughness

Ra is the arithmetic mean of the absolute deviations of the roughness profile from the centre line, as defined in ISO 4287 (16). It is the parameter reported in almost every dental paper on this subject. Contact stylus profilometry remains a standard method; optical methods add a three-dimensional map but are not required in order to rank Ra (4,16).

Bollen, Lambrechts and Quirynen set 0.2 µm as the plaque-retention threshold (1). Quirynen and Bollen had already shown that roughness and surface-free energy govern plaque accumulation on restorations (17). Jones, Billington and Pearson later asked volunteers to rank composite discs with the tongue. Discrimination lay between 0.25 µm and 0.50 µm, and they advised finishing to no more than 0.50 µm if the patient is not to feel the surface (2).

Toothbrushing raises Ra. Heintze and colleagues showed that the increase depends on material, time and load (4). Nanofilled resins, whose small particles wear with the matrix, usually remain smoother than nanohybrids, in which larger fillers are uncovered or lost (8,18). The dentifrice also matters. Monteiro and Spohr found that toothbrushing with different pastes increased roughness of composite resins, although the relationship between relative dentine abrasivity and the final Ra was not straightforward (8).

2.3 Injectable composites

Early flowables were not intended for occlusal load (13). Current highly filled injectables are a different material. G-ænial Universal Injectable is filled to about 69 wt% (50 vol%) with ultrafine barium glass and silica, mean size about 150 nm, in a UDMA / Bis-MEPP / TEGDMA matrix (6,18). Beautifil Flow Plus X F00 is a zero-flow giomer injectable containing nano-sized surface pre-reacted glass-ionomer (S-PRG) filler (19). It is not the same product as Beautifil Flow Plus F00, which appears in several earlier papers (6).

Rajabi et al. compared G-ænial Universal Injectable and Beautifil Flow Plus F00 with a conventional flowable and a nanohybrid paste in two-body wear (200 000 cycles, 49 N, steatite antagonist, dry conditions). The two injectables lost less volume than the paste and the classical flowable, and flexural strength was also higher. The two injectables did not differ from each other (6). That assay measures occlusal-contact wear as volume; it cannot be read as toothbrush mass loss.

Hasan, Eltoukhy and Zaghloul brushed G-ænial Universal Injectable, a nanofilled paste and a nanohybrid composite for times taken to represent six months and one year. After abrasion the injectable had the lowest Ra and the nanohybrid was the roughest; the two nanofilled resins did not differ (18). That design is closer to the present study.

A 2025 systematic review of highly filled flowables used as sole restoratives concluded that optical properties are generally adequate but that mechanical properties remain inferior to those of medium-viscosity composites, and advised caution in extensive cavities, load-bearing areas and patients with parafunction (20). Material-to-material scatter remains the rule, and toothbrush abrasion of Beautifil Flow Plus X F00 against Beautifil II LS was not among the comparisons reviewed.

2.4 Conventional nanohybrid resins and giomers

Nanohybrid pastes combine nano-sized particles with larger submicron or micron fillers. Total inorganic load commonly exceeds 70–80 wt% (11). Beautifil II LS is such a paste: filler load is 83 wt%, the resin uses an SRS monomer intended to reduce shrinkage, and the filler includes S-PRG glass (19). S-PRG is a three-layered filler with a fluoro-boro-aluminosilicate core, a pre-reacted glass-ionomer phase and a silica coating; it releases fluoride, strontium, borate, sodium, silicate and aluminium ions (23). The same ion-exchange behaviour can roughen the surface under acidic challenge (21). Once the matrix is brushed away, larger and irregular fillers leave a higher Ra than a uniform nanofilled injectable (18).

An eight-year clinical series of an earlier giomer paste (Beautifil with a self-etch primer) showed acceptable retention and little secondary caries in the teeth available for recall

(22). That is clinical behaviour of a related material, not toothbrush Ra of Beautifil II LS; it only establishes that S-PRG pastes have been used as definitive restoratives for years.

Packable nanohybrids remain the clinical reference for posterior direct work. They are harder to adapt in a narrow box and more likely to trap voids. Injectable products were developed to close that handling gap without sacrificing wear resistance. Whether they do so under toothbrush abrasion is the question addressed by this thesis.

2.5 What remains unclear

Two-body wear of G-ænial Universal Injectable and Beautifil Flow Plus F00 is favourable in one recent laboratory study (6). Toothbrush Ra of G-ænial Universal Injectable is favourable against a nanohybrid in another (18). Beautifil Flow Plus X F00 and Beautifil II LS have not been placed beside G-ænial Universal Injectable in the same 10 000-cycle, 2 N, gravimetric-plus-Ra protocol. That is the gap the present experiment fills.

Chapter 3

Materials and methods

3.1 Study design and sample

The study was an in-vitro comparison of three resin composites. Thirty-six discs, 10 mm in diameter and 1 mm thick, were prepared in a CAD/CAM Teflon mould and allocated at random to three groups of twelve. An initial G*Power estimate had indicated ten discs per group; twelve were used in order to increase statistical power.

3.2 Materials

1. G-ænial Universal Injectable — highly filled injectable composite, GC Corporation, Tokyo, Japan.

2. Beautifil Flow Plus X F00 — highly filled injectable giomer, Shofu Inc., Kyoto, Japan.

3. Beautifil II LS — conventional nanohybrid giomer, Shofu Inc., Kyoto, Japan.

4. Colgate Total toothpaste.

5. Distilled water.

Table 3.1 Brand, manufacturer, matrix, filler and load of the resin composites

Composite	Type	Matrix	Filler	Load	Manufacturer
G-ænial Universal Injectable	Nanofilled injectable	UDMA, Bis-MEPP, TEGDMA	Ultrafine barium glass and silica, ~150 nm	69 wt% / 50 vol %	GC, Tokyo
Beautifil Flow Plus X F00	Giomer injectable, zero flow	Dimethacrylate resin	Nano S-PRG filler	Highly filled injectable (manufacturer)	Shofu, Kyoto
Beautifil II LS	Nanohybrid giomer, low shrinkage	SRS monomer complex	S-PRG and pre-polymerised fillers	83 wt%	Shofu, Kyoto

3.3 Specimen preparation

Each material was placed in the Teflon mould. A matrix strip and glass plate were used to exclude air from the surface and to keep the disc flat. Polymerisation followed the manufacturer’s instructions for that product. Flash was removed. The discs were identified by group and stored in distilled water until testing.

3.4 Baseline measurements

Each disc was dried and weighed on an analytical balance. The recorded masses run to 0.0001 g, indicating that the balance read to 0.1 mg. Ra was measured with a contact profilometer on the test face. The value used for analysis was the mean of the traces taken on that disc.

3.5 Toothbrushing

Discs were mounted in a toothbrushing simulator. A slurry of Colgate Total and water was prepared according to ISO 11609 and ISO/TR 14569-1 guidance (3,14). The brush load was 2 N, within the ISO range of 0.5–2.5 N (3). Each disc received 10 000 cycles, the conventional laboratory equivalent of one year of twice-daily brushing (7,8). Fresh slurry was used as required to keep the abrasive present. After the test the discs were rinsed free of paste.

3.6 Post-test measurements

Mass and Ra were recorded again with the same instruments and the same drying procedure as at baseline.

Percentage weight loss = $[(W_1 - W_2) / W_1] \times 100$

where W_1 and W_2 are mass before and after brushing.

Absolute loss (mg) = $(W_1 - W_2) \times 1000$.

ΔRa (μm) = Ra after − Ra before.

Percentage weight loss was calculated as the mean of the per-disc ratios, not as the ratio of the group-mean masses.

3.7 Statistical analysis

Descriptive values are reported as mean ± standard deviation. Percentage weight loss was compared among groups by one-way ANOVA. Ra after brushing was compared by the Kruskal–Wallis test, with pairwise post-hoc comparisons. Significance was set at $\alpha = 0.05$.

Chapter 4

Results

Thirty-six discs were tested, twelve in each group. All discs completed 10 000 cycles at 2 N. Wear is reported as mass; roughness is reported as Ra.

4.1 Weight loss

Mean mass before and after brushing, and percentage loss, are given in Table 4.1 and illustrated in Figure 4.1.

Table 4.1 Mean weight before and after simulated toothbrushing and percentage weight loss of the tested composite resins (n = 12)

Composite resin	Weight before (g) Mean ± SD	Weight after (g) Mean ± SD	Weight loss (%) Mean ± SD	p-value
Beautifil Flow Plus X F00	0.1602 ± 0.0074	0.1576 ± 0.0031	1.49 ± 3.14	
G-ænial Universal Injectable	0.1630 ± 0.0102	0.1619 ± 0.0106	0.65 ± 0.37	0.329†
Beautifil II LS	0.2313 ± 0.0154	0.2303 ± 0.0156	0.41 ± 0.25	

† One-way ANOVA. Values are mean ± standard deviation.

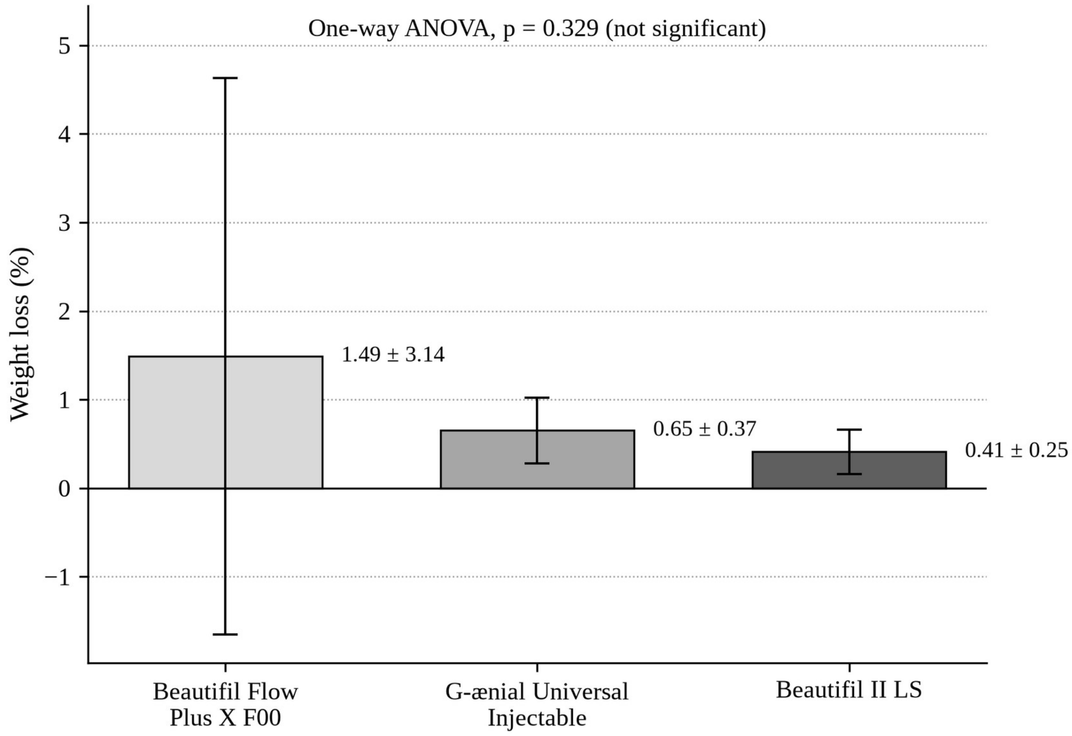

Figure 4.1 Mean percentage weight loss of the three composite resins after 10 000 toothbrushing cycles at 2 N. Error bars represent one standard deviation (n = 12 per group). In the Beautifil Flow Plus X F00 group the standard deviation exceeds the mean, so the lower error bar extends below zero; this reflects the dispersion introduced by a single specimen that

lost approximately 11.4 % of its mass and does not imply a gain in weight. Differences among groups were not statistically significant (one-way ANOVA, p = 0.329).

Beautifil II LS discs were heavier at baseline (0.2313 ± 0.0154 g) than the two injectables (0.1602 ± 0.0074 g and 0.1630 ± 0.0102 g). This follows from filler load and density, not from a difference in disc size.

One-way ANOVA on percentage loss gave p = 0.329. The first null hypothesis is therefore retained. Numerically, Beautifil II LS lost the least mass (0.41 ± 0.25 %), followed by G-ænial Universal Injectable (0.65 ± 0.37 %) and Beautifil Flow Plus X F00 (1.49 ± 3.14 %). The standard deviation in the last group is an order of magnitude larger than in the other two.

Absolute loss is set out in Table 4.2 because percentage loss on a lighter disc can exaggerate a small mass change, and because the outlying specimen is easier to see in milligrams.

Table 4.2 Absolute weight loss after 10 000 brushing cycles (n = 12)

Composite resin	**Absolute weight loss (mg) Mean ± SD**	**Range (mg)**
Beautifil Flow Plus X F00	2.58 ± 5.91	0.2 – 20.7
G-ænial Universal Injectable	1.07 ± 0.62	0.2 – 1.9
Beautifil II LS	0.95 ± 0.58	0.1 – 1.8

Values are mean ± standard deviation. The large standard deviation in Beautifil Flow Plus X F00 is attributable to one disc that lost approximately 11.4 % of its mass (20.7 mg). The remaining discs in that group, and all discs in the other two groups, lost between 0.1 mg and 1.9 mg.

The standard deviation of Beautifil Flow Plus X F00 mass fell from 7.4 mg before brushing to 3.1 mg after brushing. This is consistent with the heaviest disc in that group shedding 20.7 mg and landing near the group mean.

4.2 Surface roughness

Mean Ra before and after brushing, and ΔRa, are given in Table 4.3 and illustrated in Figures 4.2 and 4.3.

Table 4.3 Mean surface roughness (Ra) before and after simulated toothbrushing and the change in roughness (ΔRa) of the tested composite resins (n = 12)

Composite resin	Ra before (µm) Mean ± SD	Ra after (µm) Mean ± SD	ΔRa (µm) Mean ± SD	p-value
Beautifil Flow Plus X F00	0.038 ± 0.011	0.104 ± 0.017	0.066 ± 0.021	
G-ænial Universal Injectable	0.054 ± 0.010	0.100 ± 0.023	0.046 ± 0.023	< 0.001‡
Beautifil II LS	0.119 ± 0.054	0.194 ± 0.050	0.075 ± 0.066	

‡ Kruskal–Wallis test. Pairwise comparisons: each injectable versus Beautifil II LS, $p < 0.001$; the two injectables, not significant. Values are mean ± standard deviation.

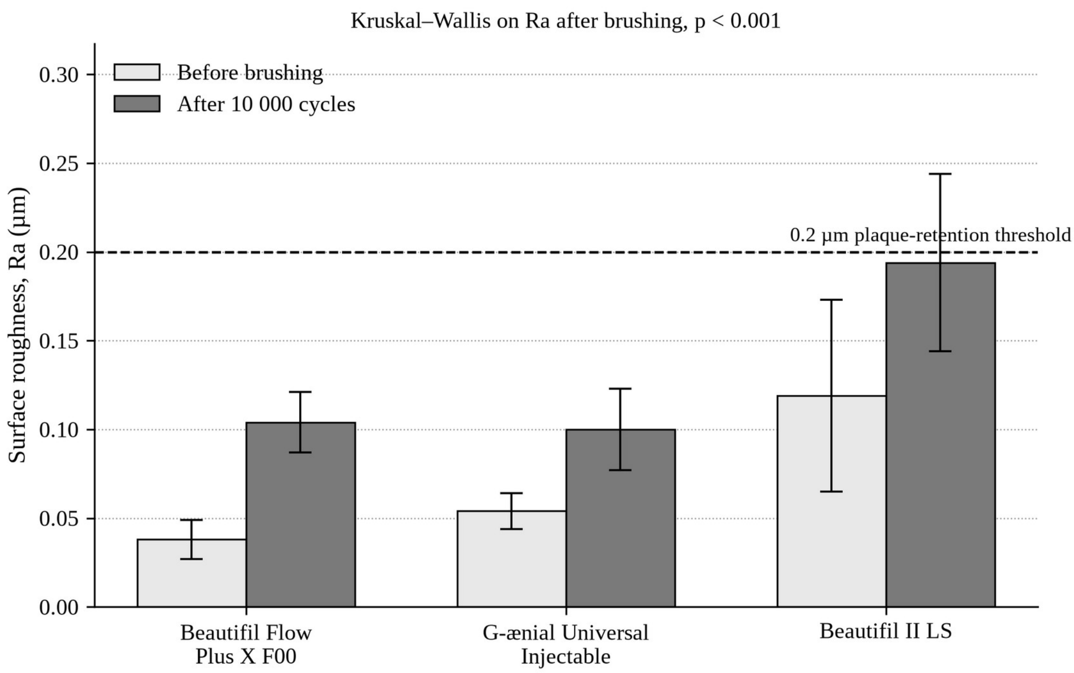

Figure 4.2 Mean surface roughness (Ra) of the three composite resins before and after 10 000 toothbrushing cycles at 2 N. Error bars represent one standard deviation (n = 12 per

group). The broken line marks the 0.2 µm threshold above which plaque retention increases. Ra after brushing differed significantly among groups (Kruskal–Wallis, $p < 0.001$).

At baseline Beautifil II LS was already the roughest (0.119 ± 0.054 µm). Beautifil Flow Plus X F00 was the smoothest (0.038 ± 0.011 µm). G-ænial Universal Injectable lay between them (0.054 ± 0.010 µm).

Every group became rougher. Mean ΔRa was 0.066 ± 0.021 µm for Beautifil Flow Plus X F00, 0.046 ± 0.023 µm for G-ænial Universal Injectable and 0.075 ± 0.066 µm for Beautifil II LS (Figure 4.3). After 10 000 cycles the means were 0.104 ± 0.017 µm, 0.100 ± 0.023 µm and 0.194 ± 0.050 µm respectively.

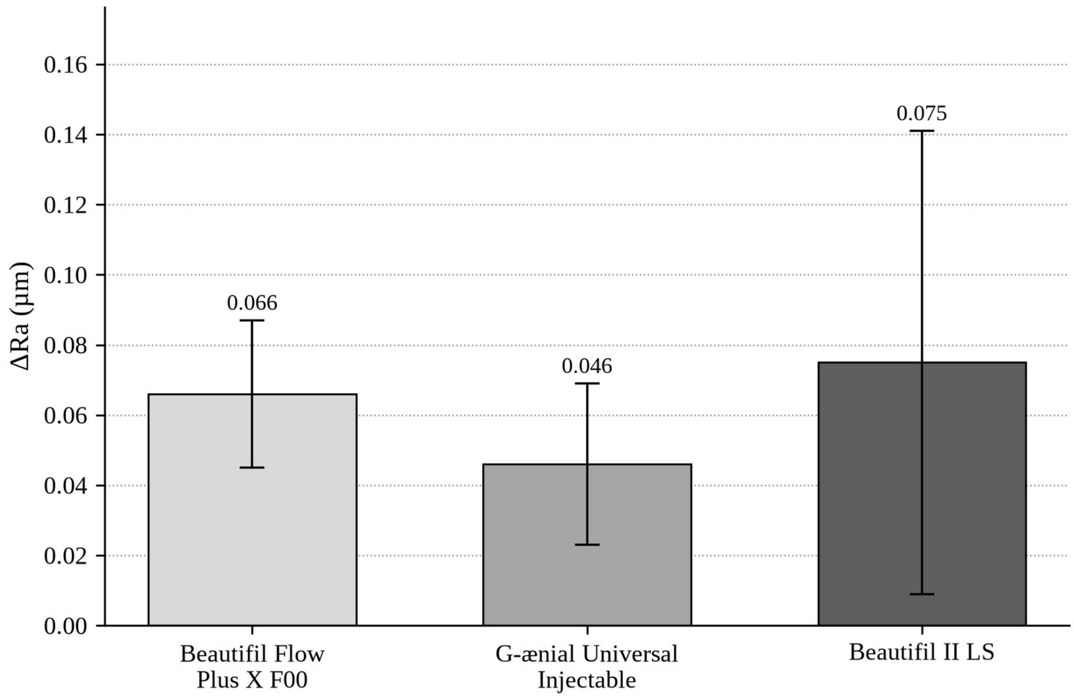

Figure 4.3 Mean change in surface roughness (ΔRa = Ra after − Ra before) of the three composite resins after 10 000 toothbrushing cycles at 2 N. Error bars represent one standard deviation ($n = 12$ per group).

The Kruskal–Wallis test on Ra after brushing was significant ($p < 0.001$). The second null hypothesis is therefore rejected. Pairwise comparisons separated Beautifil II LS from each injectable resin but did not separate the two injectables from each other.

Relative to the 0.2 µm plaque-retention threshold (1), both injectable means remained below it. The Beautifil II LS mean was 0.194 µm, on the threshold. Its standard deviation of 0.050 µm indicates that some discs in that group crossed 0.2 µm, which is visible as the upper error bar in Figure 4.2. All three means remained well below the 0.50 µm tongue-detection threshold (2).

4.3 Summary of the findings

Percentage weight loss did not differ significantly among the three resins. Absolute loss in the two well-behaved groups, and in eleven of the twelve Beautifil Flow Plus X F00 discs, was of the order of 1 mg. One disc in that group was an exception. After the same abrasion both injectable resins were significantly smoother than Beautifil II LS and did not differ from each other.

Chapter 5

Discussion

The first null hypothesis is retained. The second is rejected. The three resins lost a statistically similar fraction of their mass, but they did not finish with a similar surface roughness.

5.1 Weight loss

A non-significant ANOVA ($p = 0.329$) is not proof of equal wear; it is a failure to reject equality. Beautifil Flow Plus X F00 had a standard deviation of 3.14 % beside a mean of 1.49 %. That spread is produced by the outlying disc (approximately 11.4 %, 20.7 mg). Residual variance rises and power falls. The other eleven discs in that group, and the twenty-four discs in the other groups, lost between 0.1 mg and 1.9 mg. On that restricted view the three materials are close.

Beautifil II LS started heavier. Percentage loss on a heavier disc is a smaller number for the same milligrams lost, which is why absolute loss is the fairer comparison: 0.95 ± 0.58 mg against 1.07 ± 0.62 mg for G-ænial Universal Injectable. Those two means lie near each other. The nanofilled injectable did not lose more mass than the 83 wt% nanohybrid under this slurry.

Rajabi et al. found less volumetric two-body wear for G-ænial Universal Injectable and Beautifil Flow Plus F00 than for a nanohybrid paste (6). The direction is favourable to injectables, but the test is not the same. They used 200 000 chewing cycles, a 49 N load and a steatite ball, without toothpaste, and they measured volume. Beautifil Flow Plus F00 is also not Beautifil Flow Plus X F00. Agreement can be stated only as a general remark: highly filled injectables are not automatically the weaker partner in a wear assay.

The reason one disc lost 20.7 mg is not known. A void, an under-cured layer or a weighing error would all produce that point. The value was retained and is footnoted in Table 4.2. Any later re-analysis that excludes it must say so.

5.2 Surface roughness

The roughness result is more decisive. The nanohybrid was rougher before brushing and rougher after it. ΔRa was also largest in that group, with a wide spread (0.075 ± 0.066 µm). G-ænial Universal Injectable changed the least (0.046 ± 0.023 µm). Beautifil Flow Plus X F00 started smoothest and finished statistically indistinguishable from G-ænial Universal Injectable.

That pattern is consistent with filler geometry. A nanofilled injectable with approximately 150 nm barium glass and silica wears more evenly (6,18). A nanohybrid with larger S-PRG and pre-polymerised particles loses matrix first; the large fillers then stand proud or leave pits (4,18). Hasan, Eltoukhy and Zaghloul observed the same order after simulated brushing: G-ænial Universal Injectable smoother than a nanohybrid, and two nanofilled resins not different from each other (18). Heintze et al. had already shown that hybrid and microhybrid pastes roughen more with load and time than microfilled resins (4).

The 0.2 µm line is the clinically useful one (1,17). Both injectables ended near 0.10 µm and remain, on the mean, in the range where further smoothing is not expected to reduce plaque. Beautifil II LS ended at 0.194 ± 0.050 µm. The mean is acceptable on the published threshold, but the spread is not entirely below it. None of the means approach the 0.50 µm tongue threshold (2). Patients would not be expected to feel these surfaces after one year of this brushing model. Biofilm risk is the sharper distinction, and it applies to the nanohybrid group.

The ΔRa standard deviation of Beautifil II LS (0.066 µm) exceeds both its baseline and its final Ra standard deviations. That pattern requires a negative within-disc correlation — the roughest discs becoming relatively smoother and the smoothest becoming rougher — and is consistent with ΔRa having been computed per disc before averaging.

5.3 Clinical implications

Within this protocol the two injectable resins may be used where toothbrush abrasion is the wear of interest without paying a roughness penalty against Beautifil II LS. They do not, on these data, wear less. They finish smoother. That matters on freely cleanable surfaces and at margins. It does not license a claim that they survive occlusal contact better, because occlusal contact was not tested. The caution expressed by Tzimas et al. regarding load-bearing use of highly filled flowables therefore still stands (20).

Beautifil II LS remains a reasonable packable control. Its higher baseline Ra may reflect filler size, the finishing sequence, or both. If the nanohybrid was left with a coarser finish, part of the after-brushing difference was already present at baseline. The after-brushing comparison is nevertheless valid: the groups were compared as they stood after the same abrasion.

5.4 Strengths

The three products are current commercial materials and include the X-generation Flow Plus rather than the earlier F00 formulation. Sample size was twelve rather than ten. Both mass and Ra were recorded on the same discs. The brushing load and cycle count follow published ISO and Sexson–Phillips figures (3,7). The statistical tests match the distributions of the two outcomes. The outlying specimen was retained and identified rather than silently excluded.

5.5 Limitations

The test is in vitro. Saliva, pH cycling, enzymes and occlusal load are absent. Gravimetric loss is not worn volume and is not worn anatomy. A single dentifrice was used. Brush stiffness and slurry ratio were not fully specified. There are no scanning electron images, so filler plucking is an inference from the roughness numbers rather than a micrograph. Ten thousand cycles represent one conventional laboratory year, not a clinical year. Disc geometry is not a Class II cavity.

5.6 Concluding remarks of the discussion

After 10 000 strokes at 2 N the three resins could not be separated on percentage weight loss. They could be separated on Ra. The injectable materials were smoother. Beautifil II LS finished on the plaque-retention threshold. That is the result of this study, and it should not be enlarged into a general ranking of injectable versus nanohybrid resins outside toothbrush abrasion.

Chapter 6

Conclusions and recommendations

6.1 Conclusions

Under the conditions of this in-vitro study:

1. Percentage weight loss after 10 000 toothbrushing cycles did not differ significantly among Beautifil Flow Plus X F00, G-ænial Universal Injectable and Beautifil II LS ($p = 0.329$).

2. Ra after the same abrasion did differ ($p < 0.001$). Both injectable resins were significantly smoother than Beautifil II LS. The two injectable resins did not differ from each other.

3. Mean Ra of both injectable resins remained below 0.2 µm. Mean Ra of Beautifil II LS was 0.194 µm.

4. One Beautifil Flow Plus X F00 disc lost approximately 11.4 % of its mass and is the source of that group's large standard deviation.

5. The first null hypothesis is retained. The second is rejected.

6.2 Recommendations

Clinical practice. Where toothbrush abrasion and smoothness after hygiene are the concern, either injectable resin tested here may be considered alongside Beautifil II LS. The data do not show less wear. They show a smoother surface after this brushing model.

Further laboratory work. Repeat the mass analysis with and without the outlying disc, and report both. Add volumetric loss or profilometric wear depth. Add scanning electron microscopy of the worn surface. State the finishing sequence, the brush and the slurry ratio in full. A second dentifrice, or a higher cycle count, would test whether the roughness difference holds.

Clinical research. Only a controlled clinical study can determine whether the 0.10 µm versus 0.19 µm difference after a laboratory year changes plaque accumulation, staining or restoration survival.

Chapter 7

References

1. Bollen CML, Lambrechts P, Quirynen M. Comparison of surface roughness of oral hard materials to the threshold surface roughness for bacterial plaque retention: a review of the literature. Dent Mater. 1997;13(4):258-269.

2. Jones CS, Billington RW, Pearson GJ. The in vivo perception of roughness of restorations. Br Dent J. 2004;196(1):42-45.

3. International Organization for Standardization. ISO/TR 14569-1:2007. Dental materials — Guidance on testing of wear — Part 1: Wear by toothbrushing. Geneva: ISO; 2007.

4. Heintze SD, Forjanic M, Ohmiti K, Rousson V. Surface deterioration of dental materials after simulated toothbrushing in relation to brushing time and load. Dent Mater. 2010;26(4):306-319.

5. Bayne SC, Thompson JY, Swift EJ Jr, Stamatiades P, Wilkerson M. A characterization of first-generation flowable composites. J Am Dent Assoc. 1998;129(5):567-577.

6. Rajabi H, Denny M, Karagiannopoulos K, Petridis H. Comparison of flexural strength and wear of injectable, flowable and paste composite resins. Materials (Basel). 2024;17(19):4749.

7. Sexson JC, Phillips RW. Studies on the effects of abrasives on acrylic resins. J Prosthet Dent. 1951;1(4):454-471.

8. Monteiro B, Spohr AM. Surface roughness of composite resins after simulated toothbrushing with different dentifrices. J Int Oral Health. 2015;7(7):1-5.

9. Mair LH, Stolarski TA, Vowles RW, Lloyd CH. Wear: mechanisms, manifestations and measurement. Report of a workshop. J Dent. 1996;24(1-2):141-148.

10. Lambrechts P, Debels E, Van Landuyt K, Peumans M, Van Meerbeek B. How to simulate wear? Overview of existing methods. Dent Mater. 2006;22(8):693-701.

11. Ferracane JL. Resin composite—state of the art. Dent Mater. 2011;27(1):29-38.

12. Condon JR, Ferracane JL. In vitro wear of composite with varied cure, filler level, and filler treatment. J Dent Res. 1997;76(7):1405-1411.

13. Baroudi K, Rodrigues JC. Flowable resin composites: a systematic review and clinical considerations. J Clin Diagn Res. 2015;9(6):ZE18-ZE24.

14. International Organization for Standardization. ISO 11609:2017. Dentistry — Dentifrices — Requirements, test methods and marking. Geneva: ISO; 2017.

15. Heintze SD, Cavalleri A, Forjanic M, Zellweger G, Rousson V. A comparison of three different methods for the quantification of the in vitro wear of dental materials. Dent Mater. 2006;22(11):1051-1062.

16. International Organization for Standardization. ISO 4287:1997. Geometrical product specifications (GPS) — Surface texture: Profile method — Terms, definitions and surface texture parameters. Geneva: ISO; 1997.

17. Quirynen M, Bollen CML. The influence of surface roughness and surface-free energy on supra- and subgingival plaque formation in man. A review of the literature. J Clin Periodontol. 1995;22(1):1-14.

18. Hasan HA, Eltoukhy R, Zaghloul NM. Effect of simulated toothbrushing on surface roughness of different resin composite materials: in vitro study. Al-Azhar J Dent Sci. 2025;28. doi:10.21608/ajdsm.2023.222996.1440.

19. Shofu Inc. Beautifil II LS product information; Beautifil Flow Plus X product information. Kyoto: Shofu; [accessed 2026].

20. Tzimas K, Pappa E, Fostiropoulou M, Papazoglou E, Rahiotis C. Highly filled flowable composite resins as sole restorative materials: a systematic review. Materials (Basel). 2025;18(14):3370.

21. Yap AU, Tan SH, Wee SS, Lee CW, Lim EL, Zeng KY. Chemical degradation of composite restoratives. J Oral Rehabil. 2001;28(11):1015-1021.

22. Gordan VV, Mondragon E, Watson RE, Garvan C, Mjör IA. A clinical evaluation of a self-etching primer and a giomer restorative material: results at eight years. J Am Dent Assoc. 2007;138(5):621-627.

23. Imazato S, Nakatsuka T, Kitagawa H, Sasaki JI, Yamaguchi S. Multiple-ion releasing bioactive surface pre-reacted glass-ionomer (S-PRG) filler: innovative technology for dental treatment and care. J Funct Biomater. 2023;14(4):236.